



CLIENT SUCCESS STORY

Geysital

UX/UI case study · Mobile + IoT + Embedded

Smart geyser automation. Complete control from connectivity and live temperature through heating modes and daily routines.

React Native

BLE + Wi-Fi

MQTT

Embedded C

Overview

Geysital is a smart geyser automation system that modernizes traditional water heating by giving users complete control through a mobile app. The product connects the mobile experience to a connected geyser so people can monitor, control, and automate heating without standing at the appliance.

Core capabilities cover power, live temperature, target and range, Gas / Electric / Hybrid source selection, source priority, routines, and dual connectivity over Bluetooth Low Energy and Wi-Fi.

The Problem

Traditional geysers need manual operation and give little visibility into the heating process. Users often have to reach the appliance, check water temperature by hand, and decide when to stop heating. That is inconvenient and can waste energy.

- Manual ON/OFF operation
- Limited control over target temperature
- No convenient routine-based automation
- No convenient live temperature visibility
- Difficulty managing different energy sources
- Little feedback about current device state

Discovery

Discovery starts at BLE pairing. Users scan for the geyser, confirm the device name and ID, check connection status and signal strength, then open the dashboard. The dashboard leads with current temperature, target, power, heating state, and selected source. The example design shows 29°C current against a 45°C target, idle, with a 20°C to 75°C range.

DISCOVERY

What was missing from a traditional geyser

Manual ON/OFF at the appliance

No live temperature on a phone

Limited target temperature control

Hard to choose Gas, Electric, Hybrid

No routines around daily use

Little feedback on device state

WITHOUT THE APP

At the appliance

No live reading

No daily schedule

CONNECTED

GEYSITAL_EBCE

ID D4:E9:F4:F4:EB:CE

Nearby BLE · -32 dBm

Scan Pair Confirm ID Dashboard

29°C LIVE 45°C TARGET

Idle Element off Electric

WITH GEYSITAL

From the phone

Current and target

Around the day

Control

Visibility

Flexibility

Efficiency

Automation

Connectivity

ON/OFF

Live temp

Target + range

Source priority

Routines

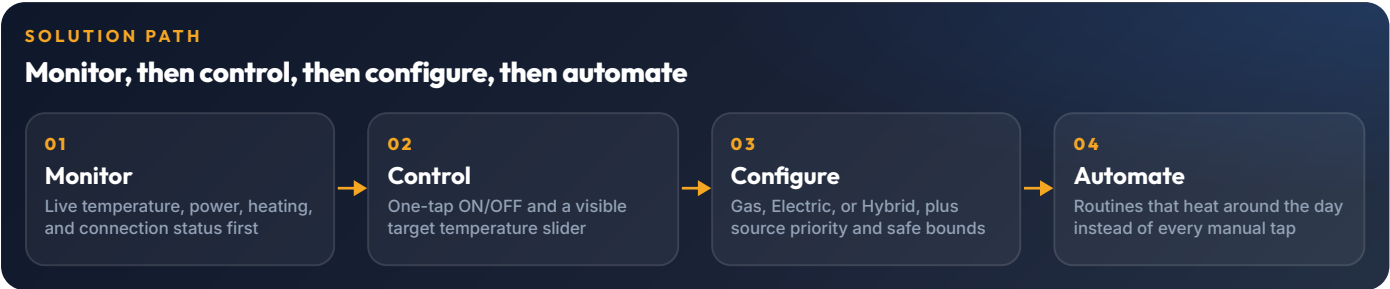
BLE + Wi-Fi

Geysital | Case Study

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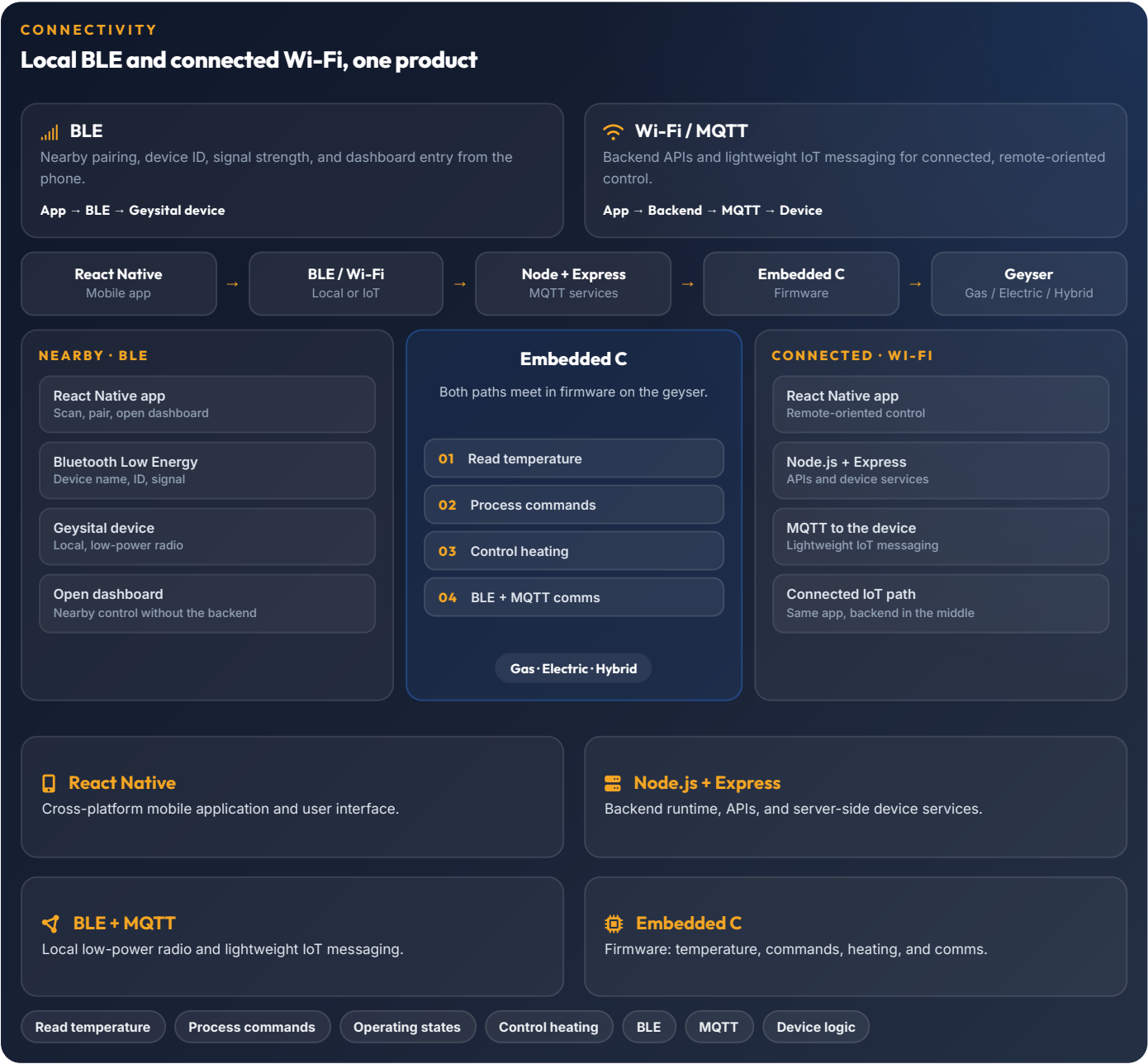
Strategy

Geysital turns the geyser into a connected smart appliance. The mobile app is the single interface for monitoring and control. Backend services, the communication layer, and Embedded C firmware operate the physical system together.



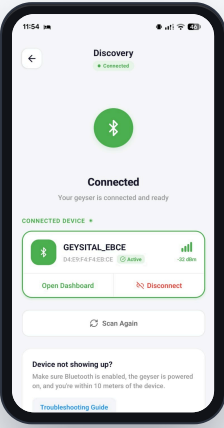
Build

Two complementary paths sit under the same app. BLE is nearby and local. Wi-Fi carries the IoT path through the backend and MQTT for remote-oriented automation. Firmware in Embedded C reads temperature, runs commands, and drives gas, electric, and hybrid hardware.



Product experience

The dashboard is the core screen. Power, live temperature, and target sit above source selection. Temperature is a primary control, not a buried setting, with a defined 20°C to 75°C operating range. Source priority lets the user prefer gas, electric, or hybrid logic.

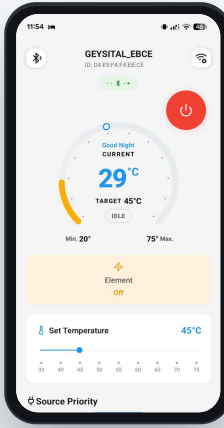


Discovery

Local BLE pairing is the entry point before the dashboard.

- Connected / disconnected status
- Device name and device ID
- BLE signal strength
- Open Dashboard or Disconnect
- Scan Again for discovery

GEYSITAL_EBCE -32 dBm Open Dashboard



Dashboard

The most important state is visible before any settings.

- Large ON/OFF power control
- Live current and target temperature
- Heating / idle and element status
- Temperature slider with safe bounds
- Gas / Electric / Hybrid priority

29°C live 45°C target Idle Electric



CONNECTED GEYSER HARDWARE

GEYSITAL_EBCE

Idle · Element off · Electric preferred

29°C
LIVE

45°C
TARGET

20 to 75°C
RANGE

ROUTINE	TIME	TARGET	MODE
Morning Shower	6:00 AM	50°C	Electric
Evening	7:00 PM	45°C	Gas
Night	10:00 PM	OFF	Standby

29°C

CURRENT EXAMPLE

45°C

TARGET EXAMPLE

20 to 75°C

SAFE OPERATING RANGE

Gas

Uses the gas heating system when that source is preferred.

Electric

Uses the electrical heating element for precise target control.

Hybrid

Uses combined or available sources according to system logic.

EXPERIENCE STRUCTURE

Clear status first, then configuration, then routines

Visual hierarchy

Power and temperature lead. Source priority and the slider sit below so status is readable before settings.

Safe bounds

The example design keeps a 20°C minimum and 75°C maximum so targets stay inside a realistic operating range.

Progressive complexity

Basic users heat water in one tap. Advanced users still reach source priority and daily routines.

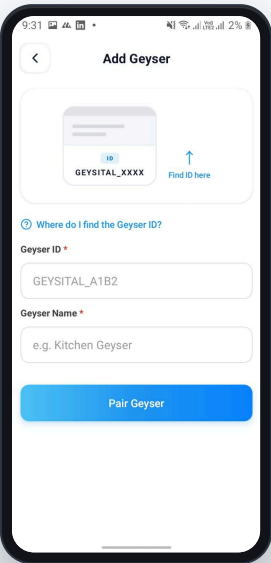
USER JOURNEY

Discover → Connect → Dashboard → Set target → Select source → Routine → Monitor

“Control your geyser. Understand its status. Choose how it heats. And let it work around your routine.”

Q Pairing and discovery

Users add a geyser by ID, then discover it over Bluetooth. Empty, scanning, and connected states stay on the same path so pairing is visible before the live dashboard.



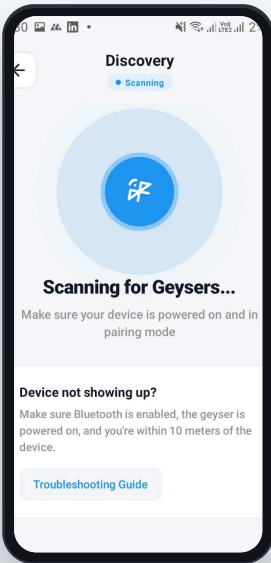
Add Geyser

Pairing starts with the ID printed on the unit and a name the household will recognize.

- Geyser ID field, required
- Geyser Name field, for example Kitchen Geyser
- Find ID here cue on the hardware label
- Pair Geyser action

GEYSITAL_XXXX Find ID here

Pair Geyser

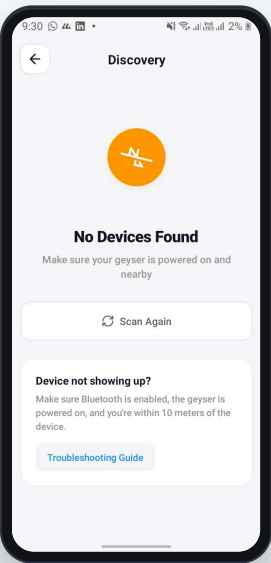


Scanning

Discovery searches nearby over BLE while the geyser stays powered on and in pairing mode.

- Scanning status on the Discovery screen
- Prompt to keep the device in pairing mode
- Help if the geyser does not appear
- Stay within 10 meters with Bluetooth on

BLE scan Pairing mode Troubleshooting

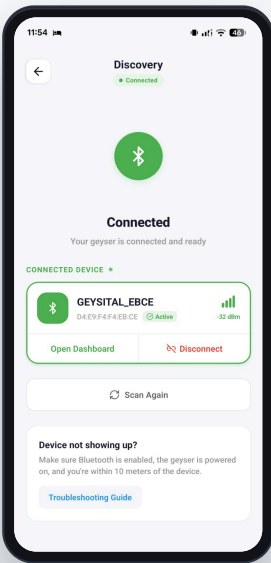


No Devices Found

When BLE does not see a unit, the empty state explains what to check instead of failing silently.

- No Devices Found status
- Scan Again to retry discovery
- Confirm the geyser is powered on and nearby
- Troubleshooting Guide for Bluetooth range

Empty state Scan Again 10 meters



Connected

A successful pair shows the device name, ID, signal, and a clear next step into the dashboard.

- GEYSITAL_EBCE marked Active
- Device ID D4:E9:F4:F4:EB:CE
- BLE signal at -32 dBm
- Open Dashboard, Disconnect, or Scan Again

Active -32 dBm Open Dashboard

PAIRING PATH

Add by ID, scan over BLE, then open the dashboard

01

Add Geyser

Enter the ID from the hardware label and a name the household will recognize.

02

Scan

Discovery searches nearby Bluetooth while the geyser is powered on and in pairing mode.

03

Empty or found

No Devices Found explains what to check. A successful pair shows GEYSITAL_EBCE as Active.

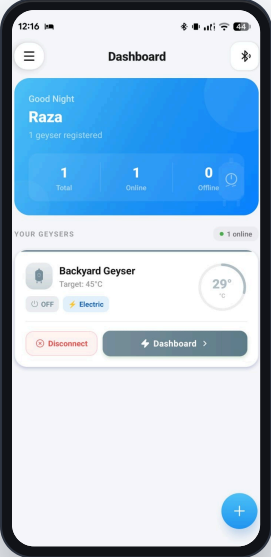
04

Open Dashboard

From the connected device card, open live control or disconnect and scan again.

📱 Home, setup, and live control

After pairing, the home list shows every registered geyser. Device setup chooses Wi-Fi or Bluetooth. The live dashboard then leads with power, temperature, and source priority.

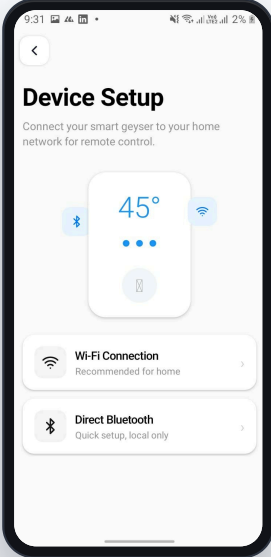


Home dashboard

The household view lists registered units, online count, and a one-tap path into each geyser.

- 1 geyser registered, 1 online, 0 offline
- Backyard Geyser at 29°C live, 45°C target
- Electric source, power OFF
- Disconnect or open Dashboard, plus add another unit

Backyard Geyser 1 online 29°C



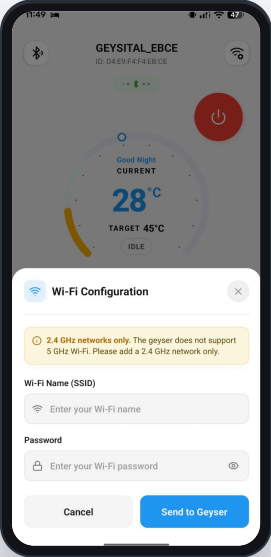
Device Setup

The same product offers two connection paths: Wi-Fi for the home network, or Bluetooth for local control.

- Wi-Fi Connection, recommended for home
- Direct Bluetooth, quick setup, local only
- Copy explains remote control over the home network
- Matches the BLE and Wi-Fi architecture

Wi-Fi Direct Bluetooth

Remote control

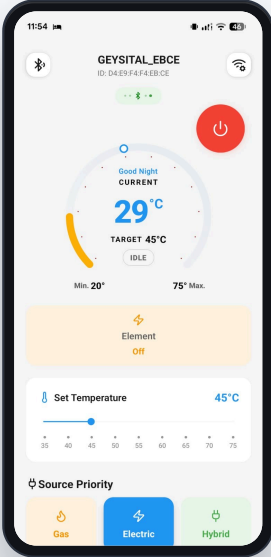


Wi-Fi Configuration

Credentials are sent to the geyser from the phone. The sheet states the radio limit before the user types an SSID.

- 2.4 GHz networks only, not 5 GHz
- Wi-Fi Name (SSID) and password
- Send to Geyser submits the network to the device
- Cancel returns to the dashboard without sending

2.4 GHz only SSID Send to Geyser



Device dashboard

Power, live temperature, and target stay above source priority so status is readable before settings.

- 29°C current against a 45°C target, idle
- Element off, with a 20°C to 75°C range
- Temperature slider as a primary control
- Gas, Electric, or Hybrid source priority

29°C live 45°C target Electric

CONTROL PATH

Household list, then network choice, then live status

01

Home dashboard

See every registered geyser, online count, live temperature, and a path into each unit.

02

Device Setup

Choose Wi-Fi Connection for the home network or Direct Bluetooth for local-only control.

03

Wi-Fi Configuration

Send a 2.4 GHz SSID and password to the geyser. 5 GHz networks are not supported.

04

Device dashboard

Power, 29°C live, 45°C target, and Gas / Electric / Hybrid sit on one screen.

🏆 Results & Impact

A traditional geyser becomes a connected appliance: React Native on the phone, Node.js and Express behind the scenes, BLE and MQTT on the wire, and Embedded C on the hardware.

